

Heyuan Huang

☎ (217)-979-7826 · ✉ heyuan@currentsurgical.com · 🌐 <https://hhuang91.github.io>
www.linkedin.com/in/heyuan-huang

SUMMARY

- 5+ years of experience in algorithm development for image reconstruction, processing, and analysis.
- Application of ML/AI in medical imaging, based on imaging physics and emphasizing reliability.
- Hands-on experience with medical imaging devices and HPC systems.

EXPERIENCE

Deep Learning Scientist, Current Surgical, Washington, DC 02/2025-now

- High-accuracy, AI-based solution for monitoring thermal acoustic ablation, combining novel neural network architecture and physics-based loss functions that improved performance by more than 30%
- High-precision 3D ultrasound navigation solution for surgical guidance, including a SOTA calibration algorithm that achieves <0.5 mm error in spatial alignment and <2 ms error in delay compensation
- UI/UX design for frontend apps, including an interactive and intuitive 2D and 3D rendering system.
- Sourcing, deploying, and maintaining the on-site high-performance machine learning workstation.

Graduate Research Assistant, Johns Hopkins University, Baltimore, MD 07/2020-02/2025

- Developed learning-based autofocus algorithm for motion artifact reduction in cone-beam CT
- Advanced efficient encoding of spatial-temporal 4D information via adaptive sampling strategy.
- Built mathematical foundation for diffusion of quantum noise medical images.
- Customized image reconstruction kernel using the Autograd framework of PyTorch
- Collected data from clinical CT systems, including scans of cadavers, phantoms, and real patients.

Research Intern, Siemens Healthineers, Princeton, NJ 06/2024-09/2024

- Delivered two prototype foundation models for AI-based deformable image registration, winning 2nd place in Best Intern Poster competition
- Designed and patented two novel network architectures for image registration tasks

Physics Intern, GE Healthcare, Troy, NY 06/2018-08/2018

- Investigated improvement in thallium doping control during production of flat panel X-ray detectors through experimentation of production conditions such as temperature, pressure, and doping duration.
- Built data management toolkit to organize, visualize, and analyze X-ray panel postproduction data for quality control and examination of potential factors affecting the production quality.

Undergraduate Researcher, University of Illinois & Carle Hospital, Urbana, IL 09/2017-09/2019

- Investigated quantitative Positron Emission Tomography (PET) protocols and algorithms.
- Published research in development of amorphous Selenium-based flat panel X-ray detectors.

EDUCATION

Ph.D. in Biomedical Engineering, Johns Hopkins University 02/2025

Research topic: Learning-based Autofocus for Motion Compensation in CT

Coursework: Human Anatomy, Machine Learning, Principles of X-ray Imaging, Medical Image Analysis, Image Processing, Information Theory, Nonlinear Optimization

BSc in Physics, University of Illinois Urbana-Champaign (UIUC) 05/2020 (GPA: 3.90/4.00)

Coursework: Classical Mechanics, Quantum Mechanics, Electrodynamics, Optics, Electronic Circuits

BSc in Nuclear, Plasma, and Radiological Engineering, UIUC 05/2020 (GPA: 3.90/4.00)

Coursework: Imaging Sensors, Radiation Interaction, Radiological Instruments, Biology, Chemistry

Journal Articles

H. Huang, Y. Liu, J.H. Siewerdsen, A. Lu, Y. Hu, W. Zbijewski, M. Unberath, C.R. Weiss, A. Sisniega, “Deformable Motion Compensation in Interventional Cone-Beam CT with a Context-Aware Learned Autofocus Metric,” *Med Phys.*, 51: 4158–4180, 2024. <https://doi.org/10.1002/mp.17125>.

H. Huang, J.H. Siewerdsen, W. Zbijewski, C. R. Weiss, M. Unberath, T. Ehtiati, A. Sisniega, “Reference-free learning-based similarity metric for motion compensation in cone-beam CT,” in *Physics in Medicine & Biology*, 67 (12), 125020, 2022. <https://doi.org/10.1088/1361-6560/ac749a>

H. Huang and S. Abbaszadeh, "Recent Development of Amorphous Selenium-Based X-ray Detectors: A Review," in *IEEE Sensors Journal*, 20 (4), 1694-1704, 2019. <https://doi.org/10.1109/JSEN.2019.2950319>

A. Lu, **H. Huang**, Y. Hu, W. Zbijewski, M. Unberath, J.H. Siewerdsen, C.R. Weiss, A. Sisniega, “Vessel-targeted compensation of deformable motion in interventional cone-beam CT,” *Medical Image Analysis*, 97, 103254, 2024. <https://doi.org/10.1016/j.media.2024.103254>

Conference Proceedings

H. Huang, A. Lu, F. Gonzalez, T. Ehtiati, J. H. Siewerdsen, A. Sisniega, “Motion Compensation for 4D Digital Subtraction Angiography via Deep Autofocus and Implicit Neural Motion Models,” in *8th International Conference on Image Formation in X-Ray Computed Tomography*, CT2024-114, accepted as oral presentation (available upon request).

H. Huang, J. H. Siewerdsen, A. Lu, Y. Hu, W. Zbijewski, M. Unberath, C. R. Weiss, and A. Sisniega, “Multi-stage Adaptive Spline Autofocus (MASA) with a learned metric for deformable motion compensation in interventional cone-beam CT”, *Proc. SPIE 12463, Medical Imaging 2023: Physics of Medical Imaging*, 1246314 (April 2023). <https://doi.org/10.1117/12.2654361>

H. Huang, J.H. Siewerdsen, W. Zbijewski, C. R. Weiss, M. Unberath, A. Sisniega, “Context-Aware, Reference-Free Local Motion Metric for CBCT Deformable Motion Compensation,” in *Proceedings of 7th International Conference on Image Formation in X-Ray Computed Tomography* (June 2022). <https://ct-meeting.org/wp-content/uploads/2022/06/CTMeetingProceedings2022.pdf>

H. Huang, J.H. Siewerdsen, W. Zbijewski, C. R. Weiss, T. Ehtiati, A. Sisniega, “Reference-Free, Learning-Based Rigid Motion Compensation in Cone-Beam CT for Interventional Neuroradiology,” in *AAPM 63rd Annual Meeting & Exhibition*. (July 2021). <https://w4.aapm.org/meetings/2021AM/programInfo/programAbs.php?sid=9204&aid=58688>

H. Huang, J.H. Siewerdsen, W. Zbijewski, C. R. Weiss, T. Ehtiati, A. Sisniega, “Reference-Free, Learning-Based Image Similarity: Application to Motion Compensation in Cone-Beam CT,” in *Proceedings of the 16th Virtual International Meeting on Fully 3D Image Reconstruction in Radiology and Nuclear Medicine* (July 2021). <https://arxiv.org/abs/2110.04143>

Patents

H. Huang, S. Reaungamornrat, F. Ghesu, and A. Kamen HEMI-INR: A Framework for Deformable Image Registration Using Instance-Agnostic Implicit Neural Representation via Hash Encoding with Multi-Resolution Inference (2025, under preparation)

S. Reaungamornrat, **H. Huang**, I. M. Popdan, F. Ghesu, and A. Kamen, Modality-Insensitive Multiresolution Foundation Model for Clinically Applicable Deformable Registration (2025, under preparation)